

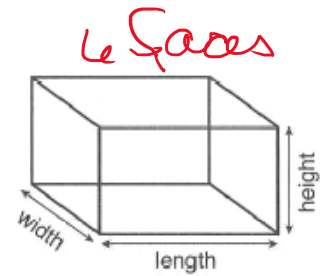
Lesson 4: Volume

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→ the space inside an object

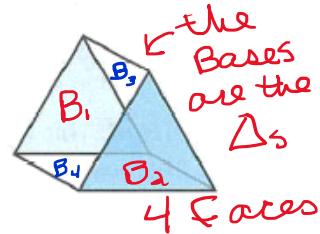
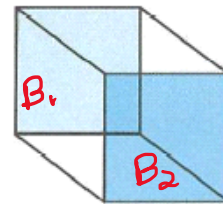
VOLUME OF PRISMS

A line has only one dimension—length. A flat shape has two dimensions—length and width. Most objects in life have three dimensions—length, width, and height. Depending on the situation, one of these dimensions may be called depth.
OR



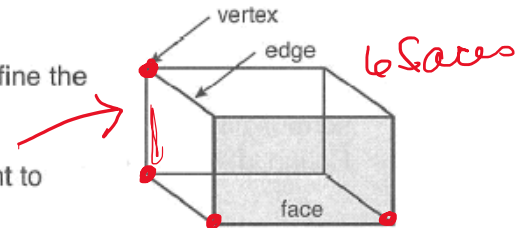
A prism is a three-dimensional object with parallel, identical bases that have straight edges. The sides, or faces, that connect the bases are always rectangles. → side

The drawing shows a rectangular prism and a triangular prism. Notice that the base determines the type of prism.



A corner of the prism is a vertex. The edges are the straight segments that define the prism's shape. The flat side of a prism is also called its face.

In the drawing, the prism has 8 vertices, 12 edges, and 6 faces. Take a moment to find them all.



Volume is the measure of the space inside a three-dimensional object. Volume is measured using cubic units. To find volume, you need to figure out how many cubes will fit inside the figure. Use the formula below to find the volume of any prism.

compare:
area is
square units

Volume (V) of a Prism

$V = Bh$, where B is the area of one base and h is the height

Examples #1

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Volume (V) of a Prism

$V = Bh$, where B is the area of one base and h is the height

each prism has a pair of parallel bases.

different h

FINDING THE VOLUME OF A PRISM

Example Find the volume of the prism.

Step 1

Identify the bases.

Remember, bases are identical and parallel. The bases are right triangles.

Step 2

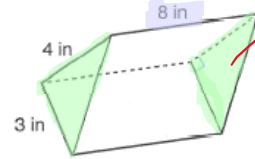
Find the area of one base.

$$A = \frac{1}{2}bh \\ = \frac{1}{2} \cdot 4 \cdot 3 \\ = 6 \text{ sq in}$$

Step 3

Multiply by the height, 8 inches.

$$6 \cdot 8 = 48 \text{ cubic inches or } 48 \text{ in}^3$$



$$A = \frac{1}{2}bh$$

$B = \text{area of one base}$

$$b = 3 \\ h = 4$$

$$B = \frac{1}{2}(3)(4)$$

$$B = 6 \text{ sq in}$$

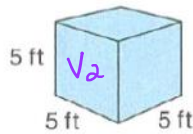
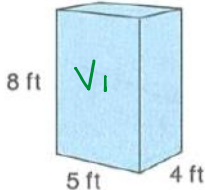
height of Prism $h = 8 \text{ in}$

$$V = Bh = 48$$

$$V = 48 \text{ cubic in}$$

7. How much greater is the volume of the first box than the volume of the second box?

$$V = Bh$$



Rectangular Prism $V = Lhw$

$$V_1 = Lwh \\ = 8(5)(4)$$

$$V_1 = 160 \text{ ft}^3$$

$$V_2 = Lwh \\ = 5^3$$

$$V_2 = 125 \text{ ft}^3$$

The 1st box is 35 cubic feet greater than the 2nd box.

How much greater is the 1st box than the 2nd box in percent?

→ how much (%) is 35 out of 125?

$$35/125 = 28\%$$

Soraiya Ahmadi 10:29 AM

why we choose 125 not 160

How much smaller is the 2nd Box than the 1st Box? $35/160$

START

8. A plan for a concrete wall calls for it to be $\frac{1}{2}$ foot thick, 6 feet high, and 48 feet long.

a. What is the volume of the wall in cubic feet?

b. Concrete is sold in cubic yards. One cubic yard = 27 cubic feet. How many cubic yards of concrete are needed to build the wall?



$$V = Lwh \\ = 48(6)(\frac{1}{2}) \\ = 48(3)$$

$$V = 144 \text{ ft}^3$$

$$\text{Change } 144 \text{ ft}^3 \text{ to } \frac{144}{27} \text{ yd}^3$$

change 144 ft^3
to yds^3

53

$$= 48(2)$$

$$V = 144 \text{ ft}^3$$

therefore
in math

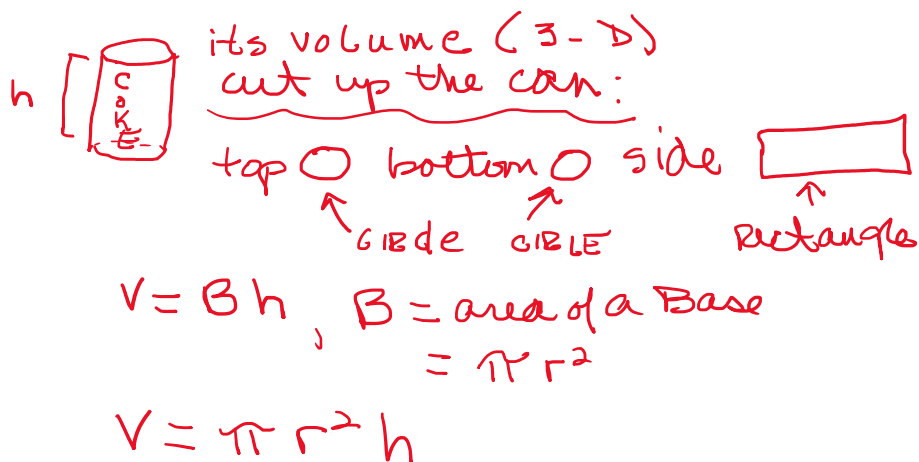
$$144 / 27 = 5.\overline{3}$$

$$\therefore 144 \text{ ft}^3 = 5.\overline{3} \text{ yd}^3$$

the correct question
how many cubic yards
of concrete must we
buy to build the wall.
 6 yds^3

Volume of Cylinders

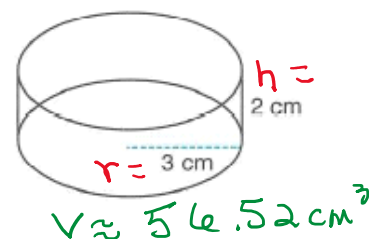
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A cylinder is a three-dimensional shape with circular bases. The two bases are identical and parallel.

To find the volume of a cylinder, find the area of one base and multiply by the height.

$$V = (\text{area of Base})(h) = \pi(3^2)(2) = 18\pi$$



The GED math formula sheet includes the formula for volume of a cylinder, which multiplies the area of one circular base (πr^2) by the height h .

Volume (V) of a Cylinder $V = \pi r^2 h$, where r = radius and h = height

Example Find the volume of a cylinder with a radius of 3 centimeters and a height of 2 centimeters.

$$\begin{aligned} V &= \pi r^2 h \\ &= 3.14 \cdot 3^2 \cdot 2 \\ &\approx 56.52 \text{ cm}^3 \end{aligned}$$

The cylinder has a volume of about **56.52 cubic centimeters** or **56.52 cm³**.

Examples #2

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9. A cylindrical fuel tank has a height of 20 feet and a diameter of 16 feet. What is the volume of the fuel tank in cubic feet?

$$V = Bh, \quad h = 20, d = 16, r = 8$$

$$B = \pi r^2$$

$$= \pi 8^2$$

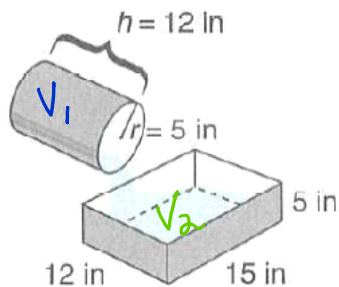
$$\approx 3.14(64)$$

$$B \approx 200.96$$

$$V \approx 200.96(20)$$

$$V \approx 4019.2 \text{ ft}^3$$

10. **Explain** The illustration shows a cylindrical can and a rectangular box. If the can in the drawing is full, is it possible to pour the contents into the box without the box overflowing?



$$V_1 = Bh$$

$$= \pi r^2 h$$

$$= \pi (5^2)(12)$$

$$V_1 \approx 942 \text{ in}^3$$

scratch

$$\begin{array}{r} 75 \\ 150 \\ \hline 75 \end{array} / 400$$

$$V_2 = Lwh$$

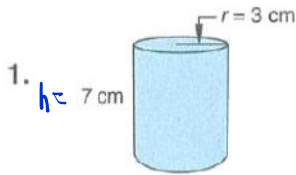
$$= 12(15)(5)$$

$$V_2 = 900 \text{ in}^3$$

Examples #3

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Find the volume of each figure. Use 3.14 for π , and round your answer to the nearest tenth if necessary.



$$\begin{aligned} V &= Bh \\ &= \pi r^2 h \\ &= \pi (3^2)(7) \\ &= 63\pi \end{aligned}$$

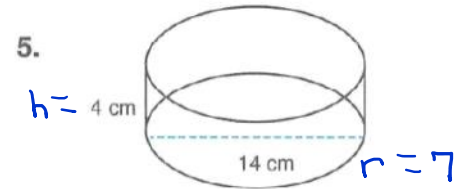
$$V \approx 197.8 \text{ cm}^3$$



(Hint: The base is a square.)

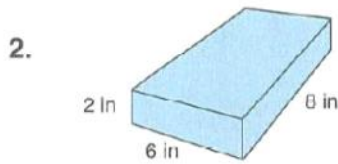
$$\begin{aligned} V &= Lwh \\ &= s^3 \\ &= 3^3 \end{aligned}$$

$$V = 27 \text{ m}^3$$



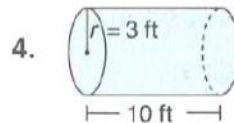
$$\begin{aligned} V &= Bh \\ &= \pi r^2 h \\ &= \pi (7^2)(4) \\ &= 196\pi \end{aligned}$$

$$V \approx 615.4 \text{ cm}^3$$



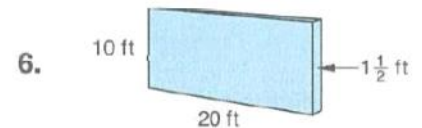
$$\begin{aligned} V &= Lwh \\ &= 2(6)(8) \end{aligned}$$

$$V = 96 \text{ in}^3$$



$$\begin{aligned} V &= Bh \\ &= \pi r^2 h \\ &= \pi (3^2)(10) \\ &= 90\pi \end{aligned}$$

$$V \approx 282.6 \text{ ft}^3$$



$$\begin{aligned} V &= Lwh \\ &= 10(20)(1.5) \\ &= 200(1.5) \end{aligned}$$

$$V = 300 \text{ ft}^3$$

VOLUME OF PYRAMIDS, CONES, AND SPHERES

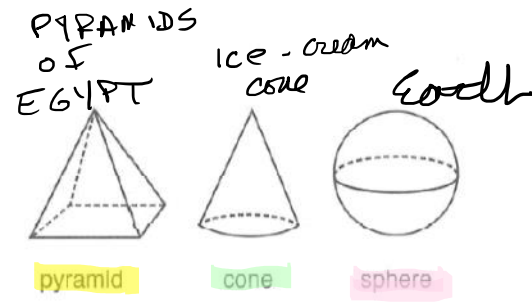
Every solid figure has volume. Remember, when you find volume, you are finding out how many cubes of a certain size would fit into the figure. Some shapes would be impossible to fill with cubes, so you need to use a formula to find the volume.

A **pyramid** has a base with straight sides and triangular faces that meet at a single vertex. The most common pyramid has a square base, but the base can be any polygon.

A **cone** has a circle for a base and one vertex.

A **sphere** is a three-dimensional object where each point on the surface is an equal distance from the center.

The GED math formula sheet has the formulas for finding the volumes of these figures. You don't need to memorize them, but take time to understand how they work.



Examples #4

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Volume (V) Formulas

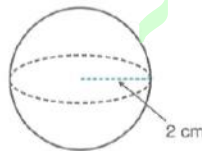
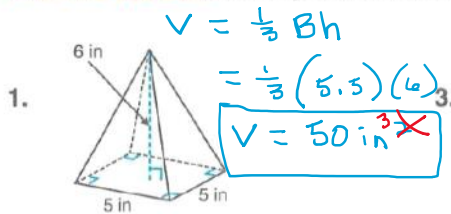
Pyramid $V = \frac{1}{3} Bh$, where B = area of the Base and h = height

Cone $V = \frac{1}{3} \pi r^2 h$, where r = radius and h = height

Sphere $V = \frac{4}{3} \pi r^3$, where r = radius

To solve volume problems, substitute the given values and solve. Use 3.14 for π , or the π key on your calculator. On drawings involving height, the height is the perpendicular distance from the base to the vertex.

rectangular prism? 3
How many pyramids fit in a
Find the volume. Round your answers to the nearest tenth.



$$V = \frac{4}{3} \pi r^3$$

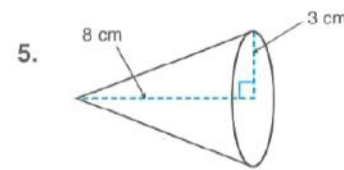
$$= \frac{4}{3} \pi (2^3)$$

$$= \frac{32}{3} \pi$$

$$V \approx 33.5 \text{ cm}^3$$

33.49
round up

How many cones in a cylinder? 3



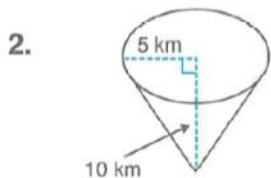
$$V = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \pi (3^2)(8)$$

$$= 24 \pi$$

$$V \approx 75.4 \text{ cm}^3$$

75.36
75.4
greater than or equal to 5

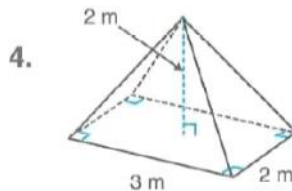


$$V = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \pi (5^2)(10)$$

$$= \frac{250}{3} \pi$$

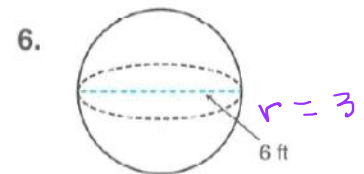
$$V \approx 261.7 \text{ km}^3$$



$$V = \frac{1}{3} Bh$$

$$= \frac{1}{3} (3)(2)(2)$$

$$V = 4 \text{ m}^3$$



$$V = \frac{4}{3} \pi r^3$$

$$= \frac{4}{3} \pi (3^3)$$

$$= 36 \pi$$

$$\approx 113.04$$

$$V \approx 113.0 \text{ ft}^3$$

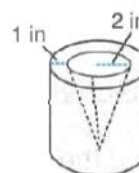
Problem Solving with Volume

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You can combine solids to form more complex figures.

FINDING THE VOLUME OF COMPLEX FIGURES

Example A cone shape is cut from a foam cylinder to create a form used to dry painted machine parts. If the height of the cylinder is 6 inches and the height of the cone is 5 inches, how many cubic inches of foam are left in the form? Round to the nearest cubic inch. (*Hint:* The radius of the cylinder is the radius of the cone plus the 1-inch thickness of the foam.)



Step 1

Find the volume of both figures.

$$\begin{aligned}\text{Cylinder: } V &= \pi r^2 h \\ &= 3.14 \cdot 3^2 \cdot 6 \\ &\approx 169.56 \text{ in}^3\end{aligned}$$

$$\begin{aligned}\text{Cone: } V &= \frac{1}{3} \pi r^2 h \\ &= \frac{1}{3} \cdot 3.14 \cdot 2^2 \cdot 5 \\ &\approx 20.93 \text{ in}^3\end{aligned}$$

There are **149 in³** left in the foam.

Step 2

Subtract and round to the nearest cubic inch.

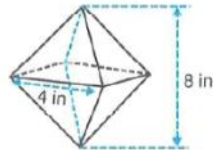
$$169.56 - 20.93 = 148.63, \text{ which rounds to } 149 \text{ in}^3.$$

Examples #5

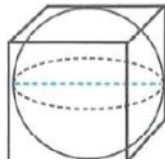
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Solve. Use 3.14 for pi.

7. A candy company combines two identical square pyramids to make a box for its premium chocolates. The drawing shows the dimensions of the box. What is the volume to the nearest cubic inch?

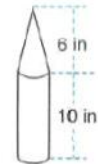


8. At a sporting goods store, basketballs are sold in boxes. Each basketball has a radius of 11.9 centimeters. Every edge of the box measures 24 centimeters. When the basketball is in the box, how much space is left in the box? Round to the nearest cubic centimeter.



9. A model rocket has the dimensions shown in the drawing. If the rocket has a diameter of 2 inches, what is the volume in cubic inches?

Round to the nearest cubic inch.



10. Max plans to build a storage shed with the dimensions shown in the drawing. What is the volume of the shed in cubic feet?

